DDES · STRUCTURAL ENGINEERING · BARRANQUILLA, COLOMBIA

Building in Colombia

A guide for foreign companies: the code, the signatures and the process, with a summary of DDES's capabilities.

To build in Colombia, a foreign company needs the same as a local one: a building permit and a structural design that meets NSR-10, signed by an engineer licensed in Colombia. If the lot allows more than 2,000 m^2, Law 1796 of 2016 also requires an independent design review and technical supervision during construction. DDES adapts your team's design to the Colombian code, signs it and answers to the licensing office.

The process, step by step

Step	Stage	What happens	Who is responsible
1	Zoning of the lot	Confirm the permitted use and the area the lot allows.	Owner and architect
2	Soil study	Required for urban and suburban buildings (NSR-10, H.1.1.2).	Geotechnical engineer
3	Structural design	Under NSR-10. A design made under another code is checked and adjusted.	Engineer licensed in Colombia
4	Independent review	If the lot allows more than 2,000 m^2. Before the permit.	Professional other than the designer
5	Building permit	Filed with the signed geotechnical study, drawings and calculations.	Urban licensing office or municipal planning office
6	Construction and technical supervision	If Law 1796 applies, a supervisor independent of the builder verifies the work.	Technical supervisor
7	Technical Certificate of Occupancy	Signed by the technical supervisor at completion. Required for occupancy.	Technical supervisor

How NSR-10 relates to your code

A design made under another code cannot be filed as it stands: it has to be checked and adjusted. These are the documents each NSR-10 title is based on, according to the code itself.

Topic	NSR-10	Base documents	What to keep in mind
Seismic	Title A	SEAOC 1999, NEHRP 2006, IBC-2009, Eurocode 8	Forces depend on the municipality's Aa and Av and on the soil profile. Usually the largest change.
Loads	Title B	ASCE/SEI 7-05, IBC-2009	Live loads from Table B.4.2.1-1 and the Colombian wind map.

Concrete	Title C	ACI 318-08	Follows ACI 318 closely, with adjustments for Colombian materials and practice.
Masonry	Title D	ACI 530-08 / ASCE 5-08 / TMS 402-08	Masonry systems commonly used in Colombia.
Steel	Title F	AISC 2010, AISC Seismic 2010	Based on AISC. A design under Eurocode or GB standards needs a full check.

Who can sign?

The drawings and calculations filed with the licensing office are signed by a professional licensed in Colombia. An engineer qualified and resident abroad can apply to COPNIA for a temporary permit (Law 842 of 2003). In practice, many foreign companies work with a local firm that adapts the design, signs it and answers for it.

Terms you will come across

Term	What it means
Curaduría urbana	A private officer who, by delegation, reviews and issues building permits in the municipalities that have one.
Law 1796 of 2016	Requires independent review and supervision when the lot allows more than 2,000 m^2.
Independent reviewer	Professional other than the designer who reviews the design before the permit.
Technical supervisor	Professional independent of the builder who verifies the structure is built as designed.
Interventoría	Technical, administrative and financial control of the works on behalf of the owner.
COPNIA	Professional council that grants engineering licenses and temporary permits to foreign engineers.

DDES in brief

Diseño de Estructuras y Soluciones S.A.S., founded in 2020 in Barranquilla, works across Colombia. It is led by Orlando Fabio Barrios Lozano, civil engineer, MSc in Civil Engineering, with more than twenty years of experience in structural calculation and site work. We design under NSR-10, ACI 318 and AISC 360, analyze in ETABS, SAP2000 and SAFE, and model in Revit and Tekla.

Project	Sector	What DDES did	Size
Edificio Di Milano	Buildings	Structural design	38,411 m^2
Edificio Malibú	Buildings	Structural design	10,000 m^2
Industrias Cannon	Industrial	Design, review and construction oversight	8,000 m^2 warehouse
Edificio Ébano	Buildings	Construction oversight and design review	6,515 m^2
Hospital Universitario CARI	Healthcare	Seismic vulnerability and rehabilitation	5,780 m^2
Bodegas Centurión	Industrial	Construction oversight, review and design	4 warehouses
Universidad de Córdoba	Education	Seismic vulnerability and rehabilitation	Montería

What we need to start

- Location of the lot and use of the project.
- Built area and number of floors.
- Existing design and the code it used, if any.
- Soil study, if it already exists.
- Target date for filing the permit.

Our team works in Spanish and reads English: you can write to us in either language. Documents filed in Colombia are in Spanish.

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General guidance based on the published text of NSR-10, Law 400 of 1997 and Law 1796 of 2016. Each project's requirements are decided by its designer and the licensing office.